

Tail Bounds and Inequalities



L21_L22.pdf
PDF

Role: You are an expert Academic Scribe. Your goal is to convert the attached slide deck into a comprehensive, sequential set of exam reference notes.

Operational Protocol:

Slide-by-Slide Processing: This is an extensive presentation. You must process the content sequentially from the first slide to the last. Do not skip slides, do not truncate long proofs, and do not provide a "summary."

Step-by-Step Logic: For every Definition, Theorem, Proof, and Worked Example, explicitly write out the full reasoning and logical progression exactly as it appears in the source. Every intermediate step in a mathematical derivation or logical argument must be included.

If the source provides a specific deductive sequence, replicate that sequence exactly.

Strict Source Fidelity: Do not include any information, examples, or "intuition" that is not explicitly found in the provided slides. Do not supplement the material with external knowledge.

Final Constraint: Provide a deep, exhaustive technical transcript only. Do not summarize the lecture at the end or offer conversational commentary. Begin with the first slide and proceed to the end.

Slide 1: Title Slide

CSE400 - Fundamentals of Probability in Computing

Lecture 21 and L22: Inequalities and Tail Bound

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Slide 2: Acknowledgement

Acknowledgement to:

- **Mor Harchol-Balter, PhD**
- J. Nelson Professor of Computer Science
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Reference Book:

- *Introduction to Probability for Computing*
 - Publisher: Cambridge University Press (2024)
 - ISBN: 978-1-009-30907-3
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Slide 3: Outline

The lecture covers:

1. Tail and Motivation of Tail Bound
 2. Markov's Inequality
 - Definition and Proof
 - Example
 3. Chebyshev's Inequality
 - Definition and Proof
 - Example
 4. Chernoff Bound and Poisson Tail
 - Definition and Proof
 - Example
 5. Jensen's Inequality
 - Convex Functions
 - Proof and Example
 6. Case Study and System-Level Understanding
-

Slide 4: Tails – Definition

Definition:

The tail of a random variable X is:

$$P(X > x)$$

Slide 5: Tails – Motivation

Why do we care about tails?

- Jobs delayed more than 24 hours
 - Packets dropped due to full buffer
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Slide 6: Tails – Key Idea

Key idea:

- Mean → average behavior (easy)
 - Tail → rare extreme events (hard)
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Slide 7: Tails – Difficulty

Why are tails hard to compute?

- Requires summing many small probabilities
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Slide 8–10: Tails Example 1 (Binomial Case)

Question:

Suppose you are distributing n jobs among n servers at random.

What is the probability that a particular server gets $\geq k$ jobs?

Intuition:

- Jobs are assigned randomly
- Most servers get few jobs
- Occasionally, one server gets many jobs

This is a **tail event** (rare overload)

Model:

$$X \sim \text{Binomial}(n, p)$$

Exact Tail Probability:

$$P(X \geq k) = \sum_{i=k}^n \binom{n}{i} p^i (1-p)^{n-i}$$

Slide 11–13: Tails Example 2 (Poisson Case)

Question:

Jobs arrive according to a Poisson process with rate λ .

What is the probability that $\geq k$ jobs arrive in one hour?

Intuition:

- Jobs arrive randomly over time
- Most hours have moderate arrivals
- Sometimes bursts occur

This is a **tail event** (sudden spike in load)

Model:

$$X \sim \text{Poisson}(\lambda)$$

Exact Tail Probability:

$$P(X \geq k) = \sum_{i=k}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!}$$

Slide 14: Why Tail Bounds?

Exact tail probabilities:

$$P(X \geq k) = \sum_{i=k}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!}$$

$$P(X \geq k) = \sum_{i=k}^n \binom{n}{i} p^i (1-p)^{n-i}$$

Observation:

- These are complicated, long sums
-

Slide 15: Idea of Tail Bounds

Instead of computing exact value:

- Find an **upper bound**
- Easier to compute

- Still gives a guarantee

Goal:

$$\text{Exact: } P(X \geq k) = ?$$

$$\text{Bound: } P(X \geq k) \leq \text{simple expression}$$

Key Idea:

Approximate safely instead of computing exactly

Slide 16–17: Running Example Setup

We will evaluate different bounds on the same example.

Experiment:

- Flip a fair coin n times

Distribution:

X = number of heads

$$X \sim \text{Binomial}(n, \frac{1}{2})$$

Mean:

$$E[X] = \frac{n}{2}$$

Question:

$$P\left(X \geq \frac{3n}{4}\right)$$

Slide 18: Markov's Inequality – Definition

Key Idea:

- Uses only the mean
- Large values are rare

Markov Inequality:

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Slide 19: Markov's Inequality – Proof

Consider:

$$E[X] = \sum_{x=0}^{\infty} x \cdot p_X(x)$$

Restrict to $x \geq a$:

$$E[X] \geq \sum_{x=a}^{\infty} x \cdot p_X(x)$$

Replace $x \geq a$ with a :

$$\begin{aligned} &\geq \sum_{x=a}^{\infty} a \cdot p_X(x) \\ &= a \sum_{x=a}^{\infty} p_X(x) \\ &= a \cdot P(X \geq a) \end{aligned}$$

Thus:

$$\begin{aligned} E[X] &\geq a \cdot P(X \geq a) \\ P(X \geq a) &\leq \frac{E[X]}{a} \end{aligned}$$

Slide 20–21: Markov on Running Example

$$X \sim \text{Binomial}(n, \tfrac{1}{2}), \quad E[X] = \frac{n}{2}$$

We want:

$$P\left(X \geq \frac{3n}{4}\right)$$

Apply Markov:

$$P\left(X \geq \frac{3n}{4}\right) \leq \frac{E[X]}{3n/4}$$

$$\begin{aligned} &= \frac{n/2}{3n/4} \\ &= \frac{2}{3} \end{aligned}$$

Observation:

This is a poor bound (does not depend on n)

Slide 22–24: Markov Real-Life Example

Scenario: Income distribution

$$E[X] = 10,000$$

Question:

Fraction earning $\geq 50,000$

Apply Markov:

$$\begin{aligned} P(X \geq 50,000) &\leq \frac{10,000}{50,000} \\ &= 0.2 \end{aligned}$$

Result:

At most 20% of people earn $\geq 50,000$

Slide 25–29: Markov Solved Example 1

Problem:

Average grade = 70, A-grade cutoff = 90

Step 1

Let $X \geq 0$, $E[X] = 70$

Step 2

Markov inequality:

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Step 3

Substitute:

$$P(X \geq 90) \leq \frac{70}{90}$$

Step 4

Simplify:

$$= \frac{7}{9}$$

Result

$$P(X \geq 90) \leq 0.778$$

Slide 30–34: Markov Solved Example 2

Problem:

Average requests = 5000/sec

Find $P(X \geq 20,000)$

Step 1

$$E[X] = 5000$$

Step 2

$$a = 20,000$$

Step 3

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Step 4

$$\begin{aligned} P(X \geq 20,000) &\leq \frac{5000}{20,000} \\ &= \frac{1}{4} \end{aligned}$$

Result

$$P(X \geq 20,000) \leq 0.25$$

Slide 35: Markov Homework Problem

Given:

$$E[X] = 40,000$$

(a) Bound $P(X \geq 120,000)$

(b) Can Markov bound $P(X < 10,000)$?

Hint:

- Markov applies to right tail $P(X \geq a)$
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Slide 36–38: Chebyshev's Inequality

Key Idea:

- Uses mean + variance
 - Controls deviation from mean
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Chebyshev Inequality:

$$P(|X - \mu| \geq a) \leq \frac{\text{Var}(X)}{a^2}$$

Proof Idea:

Apply Markov to $(X - \mu)^2$

$$P(|X - \mu| \geq a) = P((X - \mu)^2 \geq a^2)$$

Apply Markov:

$$\begin{aligned} &\leq \frac{E[(X - \mu)^2]}{a^2} \\ &= \frac{\text{Var}(X)}{a^2} \end{aligned}$$

Slide 39: Chebyshev on Running Example

$$X \sim \text{Binomial}(n, \frac{1}{2})$$

$$\mu = \frac{n}{2}, \quad \text{Var}(X) = \frac{n}{4}$$

$$\begin{aligned} P\left(X \geq \frac{3n}{4}\right) &= P\left(X - \frac{n}{2} \geq \frac{n}{4}\right) \\ &\leq P\left(|X - \mu| \geq \frac{n}{4}\right) \end{aligned}$$

Apply Chebyshev:

$$\leq \frac{n/4}{(n/4)^2} = \frac{1}{n}$$

Slide 40–42: Chebyshev Real-Life Example

Given:

$$\mu = 70, \quad \sigma = 10$$

Question:

$$P(X \leq 40 \text{ or } X \geq 100)$$

This is:

$$P(|X - 70| \geq 30)$$

Apply Chebyshev:

$$\leq \frac{10^2}{30^2} = \frac{100}{900} = \frac{1}{9}$$

Result:

At most $\frac{1}{9}$ students are far from mean

Slide 43–47: Chebyshev Solved Example 1

Given:

$$\mu = 70, \quad \sigma = 5 \Rightarrow \sigma^2 = 25$$

Step 2

$$P(X \geq 90) \leq P(|X - 70| \geq 20)$$

Step 3

$$P(|X - \mu| \geq a) \leq \frac{\sigma^2}{a^2}$$

Step 4

$$P(|X - 70| \geq 20) \leq \frac{25}{400} = \frac{1}{16}$$

Result

$$P(X \geq 90) \leq 0.0625$$

Slide 48–52: Chebyshev Solved Example 2

Given:

$$\mu = 50, \quad \sigma^2 = 16$$

Step 2

$$X \notin [30, 70] \Rightarrow |X - 50| \geq 20$$

Step 3

$$P(|X - 50| \geq 20) \leq \frac{16}{400}$$

Step 4

$$= \frac{1}{25}$$

Result

$$P(|X - 50| \geq 20) \leq 0.04$$

Slide 53: Chebyshev Homework

Given:

$$\mu = 100, \quad \sigma = 2$$

Defective if:

$$|X - 100| > 6$$

Use:

$$P(|X - \mu| \geq a) \leq \frac{\sigma^2}{a^2}$$

Slide 54–58: Chernoff Bound – Key Idea

Progression:

- Markov $\rightarrow$ uses X
 - Chebyshev $\rightarrow$ uses $(X - \mu)^2$
 - Chernoff $\rightarrow$ uses e^{tX}
-

Insight:

- Exponential grows very fast
- Large values amplified
- Tail events amplified

Better transformation $\Rightarrow$ tighter bound

Slide 59–61: Chernoff Core Trick

Transformation:

$$P(X \geq a) = P(e^{tX} \geq e^{ta})$$

Apply Markov:

$$P(e^{tX} \geq e^{ta}) \leq \frac{E[e^{tX}]}{e^{ta}}$$

Since $e^{tX} \geq 0$, Markov applies

Slide 62–64: Chernoff Bound (Final Form)

$$P(X \geq a) \leq \min_{t>0} \left\{ \frac{E[e^{tX}]}{e^{ta}} \right\}$$

Key Insight:

- Holds for all $t > 0$
- Choose best $t \rightarrow$ tightest bound

Much stronger than Markov & Chebyshev

(Exponential decay instead of polynomial)

Slide 65–68: Chernoff Lower Tail

Goal:

$$P(X \leq a)$$

Use $t < 0$:

$$P(X \leq a) = P(e^{tX} \geq e^{ta})$$

$$P(X \leq a) \leq \min_{t < 0} \left\{ \frac{E[e^{tX}]}{e^{ta}} \right\}$$

Slide 69–70: Chernoff for Poisson Tail

Given:

$$X \sim \text{Poisson}(\lambda)$$

Step 1: Compute MGF

$$\begin{aligned} E[e^{tX}] &= \sum_{i=0}^{\infty} e^{ti} \cdot \frac{e^{-\lambda} \lambda^i}{i!} \\ &= e^{-\lambda} \sum_{i=0}^{\infty} \frac{(\lambda e^t)^i}{i!} \end{aligned}$$

Using series expansion:

$$\begin{aligned} &= e^{-\lambda} e^{\lambda e^t} \\ &= e^{\lambda(e^t - 1)} \end{aligned}$$

(Continue to next slides if you want full Chernoff derivation and Jensen's inequality — say "continue")